

Figure 1.6 This spectacular mishap was the result of a very small percent error—traveling a few meters too far at the end of a journey of hundreds of thousands of meters.



TABLE 1.2 Using Significant Figures

Multiplication or division:

Result can have no more significant figures than the factor with the fewest significant figures:

$$\frac{0.745 \times 2.2}{3.885} = 0.42$$

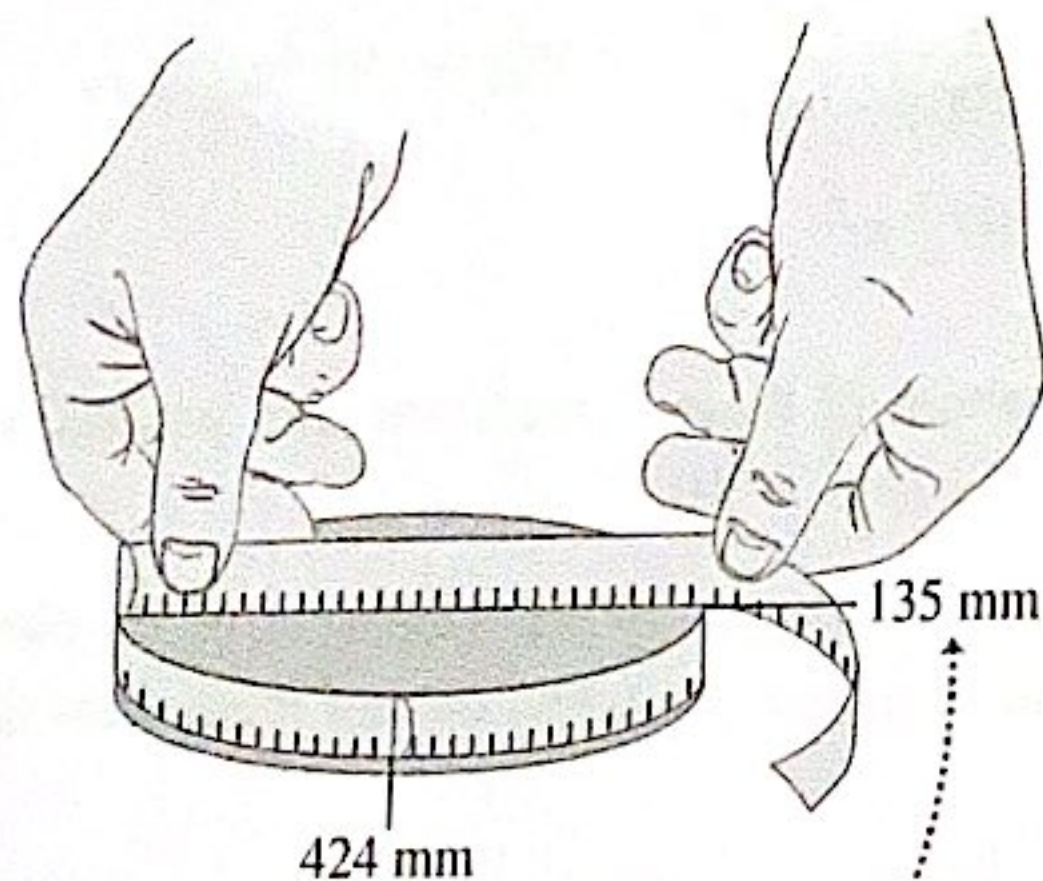
$$1.32578 \times 10^7 \times 4.11 \times 10^{-3} = 5.45 \times 10^4$$

Addition or subtraction:

Number of significant figures is determined by the term with the largest uncertainty (i.e., fewest digits to the right of the decimal point):

$$27.153 + 138.2 - 11.74 = 153.6$$

Figure 1.7 Determining the value of π from the circumference and diameter of a circle.



The measured values have only three significant figures, so their calculated ratio (π) also has only three significant figures.

1.5 UNCERTAINTY AND SIGNIFICANT FIGURES

Measurements always have uncertainties. If you measure the thickness of the cover of a hardbound version of this book using an ordinary ruler, your measurement is reliable to only the nearest millimeter, and your result will be 3 mm. It would be *wrong* to state this result as 3.00 mm; given the limitations of the measuring device, you can't tell whether the actual thickness is 3.00 mm, 2.85 mm, or 3.11 mm. But if you use a micrometer caliper, a device that measures distances reliably to the nearest 0.01 mm, the result will be 2.91 mm. The distinction between the measurements with a ruler and with a caliper is in their **uncertainty**; the measurement with a caliper has a smaller uncertainty. The uncertainty is also called the **error** because it indicates the maximum difference there is likely to be between the measured value and the true value. The uncertainty or error of a measured value depends on the measurement technique used.

We often indicate the **accuracy** of a measured value—that is, how close it is likely to be to the true value—by writing the number, the symbol $\pm$, and a second number indicating the uncertainty of the measurement. If the diameter of a steel rod is given as 56.47 ± 0.02 mm, this means that the true value is likely to be within the range from 56.45 mm to 56.49 mm. In a commonly used shorthand notation, the number 1.6454(21) means 1.6454 ± 0.0021 . The numbers in parentheses show the uncertainty in the final digits of the main number.

We can also express accuracy in terms of the maximum likely **fractional error** or **percent error** (also called *fractional uncertainty* and *percent uncertainty*). A resistor labeled “47 ohms $\pm 10\%$ ” probably has a true resistance that differs from 47 ohms by no more than 10% of 47 ohms—that is, by about 5 ohms. The resistance is probably between 42 and 52 ohms. For the diameter of the steel rod given above, the fractional error is $(0.02 \text{ mm})/(56.47 \text{ mm})$, or about 0.0004; the percent error is $(0.0004)(100\%)$, or about 0.04%. Even small percent errors can be very significant (Fig. 1.6).

In many cases the uncertainty of a number is not stated explicitly. Instead, the uncertainty is indicated by the number of meaningful digits, or **significant figures**, in the measured value. We gave the thickness of the cover of the book as 2.91 mm, which has three significant figures. By this we mean that the first two digits are known to be correct, while the third digit is uncertain. The last digit is in the hundredths place, so the uncertainty is about 0.01 mm. Two values with the *same* number of significant figures may have *different* uncertainties; a distance given as 137 km also has three significant figures, but the uncertainty is about 1 km. A distance given as 0.25 km has two significant figures (the zero to the left of the decimal point doesn't count); if given as 0.250 km, it has three significant figures.

When you use numbers that have uncertainties to compute other numbers, the computed numbers are also uncertain. When numbers are multiplied or divided, the result can have no more significant figures than the factor with the fewest significant figures has. For example, $3.1416 \times 2.34 \times 0.58 = 4.3$. When we add and subtract numbers, it's the location of the decimal point that matters, not the number of significant figures. For example, $123.62 + 8.9 = 132.5$. Although 123.62 has an uncertainty of about 0.01, 8.9 has an uncertainty of about 0.1. So their sum has an uncertainty of about 0.1 and should be written as 132.5, not 132.52. Table 1.2 summarizes these rules for significant figures.

To apply these ideas, suppose you want to verify the value of π , the ratio of the circumference of a circle to its diameter. The true value of this ratio to ten digits is 3.141592654. To test this, you draw a large circle and measure its circumference and diameter to the nearest millimeter, obtaining the values 424 mm and 135 mm (Fig. 1.7). You enter these into your calculator and obtain the quotient $(424 \text{ mm})/(135 \text{ mm}) = 3.140740741$. This may seem to disagree with the true value of π , but keep in mind that each of your measurements has three significant figures, so your measured value of π can have only three significant figures. It should be stated simply as 3.14. Within the limit of three significant figures, your value does agree with the true value.

In the examples and problems in this book we usually give numerical values with three significant figures, so your answers should usually have no more than three significant figures. (Many numbers in the real world have even less accuracy. The speedometer in a car, for example, usually gives only two significant figures.) Even if you do the arithmetic with a

calculator that displays ten digits, a ten-digit answer would misrepresent the accuracy of the results. Always round your final answer to keep only the correct number of significant figures or, in doubtful cases, one more at most. In Example 1.1 it would have been wrong to state the answer as 341.01861 m/s. Note that when you reduce such an answer to the appropriate number of significant figures, you must *round*, not *truncate*. Your calculator will tell you that the ratio of 525 m to 311 m is 1.688102894; to three significant figures, this is 1.69, not 1.68.

Here's a special note about calculations that involve multiple steps: As you work, it's helpful to keep extra significant figures in your calculations. Once you have your final answer, round it to the correct number of significant figures. This will give you the most accurate results.

When we work with very large or very small numbers, we can show significant figures much more easily by using **scientific notation**, sometimes called **powers-of-10 notation**. The distance from the earth to the moon is about 384,000,000 m, but writing the number in this form doesn't indicate the number of significant figures. Instead, we move the decimal point eight places to the left (corresponding to dividing by 10^8) and multiply by 10^8 ; that is,

$$384,000,000 \text{ m} = 3.84 \times 10^8 \text{ m}$$

In this form, it is clear that we have three significant figures. The number 4.00×10^{-7} also has three significant figures, even though two of them are zeros. Note that in scientific notation the usual practice is to express the quantity as a number between 1 and 10 multiplied by the appropriate power of 10.

When an integer or a fraction occurs in an algebraic equation, we treat that number as having no uncertainty at all. For example, in the equation $v_x^2 = v_{0x}^2 + 2a_x(x - x_0)$, which is Eq. (2.13) in Chapter 2, the coefficient 2 is *exactly* 2. We can consider this coefficient as having an infinite number of significant figures (2.000000...). The same is true of the exponent 2 in v_x^2 and v_{0x}^2 .

Finally, let's note that **precision** is not the same as **accuracy**. A cheap digital watch that gives the time as 10:35:17 a.m. is very *precise* (the time is given to the second), but if the watch runs several minutes slow, then this value isn't very *accurate*. On the other hand, a grandfather clock might be very accurate (that is, display the correct time), but if the clock has no second hand, it isn't very precise. A high-quality measurement is both *precise and accurate*.

EXAMPLE 1.3 Significant figures in multiplication

The rest energy E of an object with rest mass m is given by Albert Einstein's famous equation $E = mc^2$, where c is the speed of light in vacuum. Find E for an electron for which (to three significant figures) $m = 9.11 \times 10^{-31} \text{ kg}$. The SI unit for E is the joule (J); $1 \text{ J} = 1 \text{ kg} \cdot \text{m}^2/\text{s}^2$.

IDENTIFY and SET UP Our target variable is the energy E . We are given the value of the mass m ; from Section 1.3 (or Appendix G) the speed of light is $c = 2.99792458 \times 10^8 \text{ m/s}$.

EXECUTE Substituting the values of m and c into Einstein's equation, we find

$$\begin{aligned} E &= (9.11 \times 10^{-31} \text{ kg})(2.99792458 \times 10^8 \text{ m/s})^2 \\ &= (9.11)(2.99792458)^2(10^{-31})(10^8)^2 \text{ kg} \cdot \text{m}^2/\text{s}^2 \\ &= (81.87659678)(10^{[-31+(2 \times 8)]}) \text{ kg} \cdot \text{m}^2/\text{s}^2 \\ &= 8.187659678 \times 10^{-14} \text{ kg} \cdot \text{m}^2/\text{s}^2 \end{aligned}$$

Since the value of m was given to only three significant figures, we must round this to

$$E = 8.19 \times 10^{-14} \text{ kg} \cdot \text{m}^2/\text{s}^2 = 8.19 \times 10^{-14} \text{ J}$$

EVALUATE While the rest energy contained in an electron may seem ridiculously small, on the atomic scale it is tremendous. Compare our answer to 10^{-19} J , the energy gained or lost by a single atom during a typical chemical reaction. The rest energy of an electron is about 1,000,000 times larger! (We'll discuss the significance of rest energy in Chapter 37.)

KEYCONCEPT When you are multiplying (or dividing) quantities, the result can have no more significant figures than the quantity with the fewest significant figures.

TEST YOUR UNDERSTANDING OF SECTION 1.5 The density of a material is equal to its mass divided by its volume. What is the density (in kg/m^3) of a rock of mass 1.80 kg and volume $6.0 \times 10^{-4} \text{ m}^3$? (i) $3 \times 10^3 \text{ kg}/\text{m}^3$; (ii) $3.0 \times 10^3 \text{ kg}/\text{m}^3$; (iii) $3.00 \times 10^3 \text{ kg}/\text{m}^3$; (iv) $3.000 \times 10^3 \text{ kg}/\text{m}^3$; (v) any of these—all of these answers are mathematically equivalent.

ANSWER

number with the fewest significant figures controls the number of significant figures in the result. **|| (iii) Density = $(1.80 \text{ kg})/(6.0 \times 10^{-4} \text{ m}^3) = 3.0 \times 10^3 \text{ kg}/\text{m}^3$. When we multiply or divide, the**

1.6 ESTIMATES AND ORDERS OF MAGNITUDE

We have stressed the importance of knowing the accuracy of numbers that represent physical quantities. But even a very crude estimate of a quantity often gives us useful information. Sometimes we know how to calculate a certain quantity, but we have to guess at the data we need for the calculation. Or the calculation might be too complicated to carry out exactly, so we make rough approximations. In either case our result is also a guess, but such a guess can be useful even if it is uncertain by a factor of two, ten, or more. Such calculations are called **order-of-magnitude estimates**. The great Italian-American nuclear physicist Enrico Fermi (1901–1954) called them “back-of-the-envelope calculations.”

Exercises 1.15 through 1.20 at the end of this chapter are of the estimating, or order-of-magnitude, variety. Most require guesswork for the needed input data. Don't try to look up a lot of data; make the best guesses you can. Even when they are off by a factor of ten, the results can be useful and interesting.

EXAMPLE 1.4 An order-of-magnitude estimate

You are writing an adventure novel in which the hero escapes with a billion dollars' worth of gold in his suitcase. Could anyone carry that much gold? Would it fit in a suitcase?

IDENTIFY, SET UP, and EXECUTE Gold sells for about \$40 a gram (the price per gram has varied between \$34 and \$45 over the past five years or so), or about \$1000 for 25 grams, that is about \$1 million for 25 kilograms. A billion (1×10^9) dollars' worth of gold has a mass 10^3 times greater, about 25,000 kilograms or 25 tonnes! No human could lift it, let alone carry it. (25 tonnes is about the same as the mass of five asian elephants or two double-decker buses.)

What would the density of gold need to be in order for this amount to fit in a suitcase? The same amount of water would have the volume of 25 m^3 or 25,000 liters (the density of water is $1 \text{ g/cm}^3 = 1000 \text{ kg/m}^3$ and $1 \text{ liter} = 1 \text{ L} = 10^{-3} \text{ m}^3$). This is more than 100 times the capacity of even the largest suitcase (120 to 160 L). Therefore, for 25,000 kilograms of gold to fit in a suitcase, gold would need to be at least 100 times denser than

water. In other words, a cube of gold of side 10 cm would have a mass larger than 100 kg, which is certainly not the case. Gold is much denser than water, but not that dense. (The density of gold is actually 19.3 g/cm^3 , which is roughly 20 times that of water; the densest naturally occurring element on earth is osmium which has the density of 22.6 g/cm^3 .)

EVALUATE Clearly your novel needs rewriting. Maybe your hero could be satisfied with 1 million dollars' worth of gold? We have seen that the mass of gold in this case is about 25 kilograms, an amount which your hero should be able to carry and which would easily fit in a briefcase. If you want a more spectacular amount, try the calculation again with a suitcase full of five-carat (1-gram) diamonds, each worth \$500,000. Would this work?

KEYCONCEPT To decide whether the numerical value of a quantity is reasonable, assess the quantity in terms of other quantities that you can estimate, even if only roughly.

TEST YOUR UNDERSTANDING OF SECTION 1.6 Can you estimate the total number of teeth in the mouths of all the students on your campus? (*Hint:* How many teeth are in your mouth? Count them!)

ANSWER

The answer depends on how many students are enrolled at your campus.

1.7 VECTORS AND VECTOR ADDITION

Some physical quantities, such as time, temperature, mass, and density, can be described completely by a single number with a unit. But many other important quantities in physics have a *direction* associated with them and cannot be described by a single number. A simple example is the motion of an airplane: We must say not only how fast the plane is moving but also in what direction. The speed of the airplane combined with its direction of motion constitute a quantity called *velocity*. Another example is *force*, which in physics means a push or pull exerted on an object. Giving a complete description of a force means describing both how hard the force pushes or pulls on the object and the direction of the push or pull.

When a physical quantity is described by a single number, we call it a **scalar quantity**. In contrast, a **vector quantity** has both a **magnitude** (the “how much” or “how big” part) and a direction in space. Calculations that combine scalar quantities use the operations of ordinary arithmetic. For example, $6 \text{ kg} + 3 \text{ kg} = 9 \text{ kg}$, or $4 \times 2 \text{ s} = 8 \text{ s}$. However, combining vectors requires a different set of operations.

To understand more about vectors and how they combine, we start with the simplest vector quantity, **displacement**. Displacement is a change in the position of an object.

Displacement is a vector quantity because we must say not only how far but also in what direction. Walking 3 km north from the same place as walking 3 km southeast; these two displacements are the same but different directions.

We usually represent a vector quantity such as displacement by an arrow, as shown in Fig. 1.8a. In this book we always print vector symbols in **boldface** and put an arrow above them. We do this to remind you that vectors are different from scalar quantities; the arrow is a reminder to you to always write it with an arrow to distinguish between scalar and vector quantities in your work. To avoid the distinction in your thinking either, and confusion.

We always *draw* a vector as a line with an arrow at its end. The length of the arrow shows the vector's magnitude, and the direction of the arrow shows the vector's direction. Displacement is always a straight-line segment from the starting point to the ending point, even though the object's actual path may be curved. For example, if you go from point P_1 to point P_2 and then return to P_1 , the displacement for the trip is zero.

If two vectors have the same direction, they are *parallel*. If two vectors have the same magnitude and the same direction, they are *equal*, no matter where they are. The vector $\vec{A}'$ from point P_3 to point P_4 in Fig. 1.9 is equal to the vector $\vec{A}$ from P_1 to P_2 . These two displacement vectors are equal even though they are at different points. We write this as $\vec{A}' = \vec{A}$ in Fig. 1.9. This equality of two vector quantities is not the same as the equality of two scalar quantities. Two vector quantities are equal only when they have the same direction.

Vector $\vec{B}$ in Fig. 1.9, however, is not equal to $\vec{A}$. Vector $\vec{B}$ has the same magnitude as $\vec{A}$ but the opposite direction. We define the **negative of a vector** as a vector that has the same magnitude as the original vector but the opposite direction. The negative of a vector $\vec{A}$ is written $-\vec{A}$, and we use a boldface minus sign to emphasize the negative. If $\vec{A}$ is 87 m south, then $-\vec{A}$ is 87 m north. Thus $\vec{A} = -(-\vec{A})$ and $-\vec{A} = -(-\vec{A})$. When we write $\vec{A} = -\vec{B}$ or $\vec{B} = -\vec{A}$, we mean that the two vectors have the same magnitude and opposite directions, whether their magnitudes are the same or not.

We usually represent the *magnitude* of a vector quantity by a scalar quantity, the magnitude of the vector, but in *lightface italic type* with *no arrow* above it. If the magnitude of vector $\vec{A}$ is 87 m south, then $A = 87 \text{ m}$. An alternative way to write the magnitude of a vector is $|\vec{A}|$.

$$(\text{Magnitude of } \vec{A}) = |\vec{A}|$$

The magnitude of a vector quantity is a scalar quantity. Note that a vector can never be equal to a scalar because they are different types of quantities. The expression “ $\vec{A} = 6 \text{ m}$ ” is just as wrong as “ $6 \text{ m} = \vec{A}$ ”.

When we draw diagrams with vectors, it's best to use a scale. For example, a displacement of 5 km might be represented by a vector 1 cm long, and a displacement of 10 km by a vector 2 cm long.

Vector Addition and Subtraction

Suppose a particle undergoes a displacement $\vec{A}$ followed by a displacement $\vec{B}$. The final result is the same as if the particle had started at the same point and undergone a single displacement $\vec{C}$ (Fig. 1.10a, next page). We call $\vec{C}$ the **resultant**, or **resultant**, of displacements $\vec{A}$ and $\vec{B}$. We express this as

$$\vec{C} = \vec{A} + \vec{B}$$

The boldface plus sign emphasizes that adding two vectors is a vector process and is not the same operation as adding two scalars. For example, $2 + 3 = 5$. In vector addition we usually place the tail of the second vector at the tip of the first vector (Fig. 1.10a).

APPLICATION Scalar

Temperature, Vector Wind The comfort level on a wintry day depends on the temperature, a scalar quantity that can be positive or negative (say, $+5^\circ\text{C}$ or -20°C) but has no direction. It also depends on the wind velocity, a vector quantity with both magnitude and direction (for example, 15 km/h from the west).



1.7 • What will be your age after 0.1 billion minutes from now if you are 18 year old? (Assume 365 days per year)

1.8 • BIO (a) The recommended daily allowance (RDA) of the trace metal magnesium is 410 mg/day for males. Express this quantity in $\mu\text{g/day}$. (b) For adults, the RDA of the amino acid lysine is 12 mg per kg of body weight. How many grams per day should a 77-kg adult receive? (c) A typical multivitamin tablet can contain 2.0 mg of vitamin B₂ (riboflavin), and the RDA is 0.0030 g/day. How many such tablets should a person take each day to get the proper amount of this vitamin, if he gets none from other sources? (d) The RDA for the trace element selenium is 0.000070 g/day. Express this dose in mg/day.

1.9 •• Neptunium. In the fall of 2002, scientists at Los Alamos National Laboratory determined that the critical mass of neptunium-237 is about 60 kg. The critical mass of a fissionable material is the minimum amount that must be brought together to start a nuclear chain reaction. Neptunium-237 has a density of 19.5 g/cm^3 . What would be the radius of a sphere of this material that has a critical mass?

1.10 •• BIO Bacteria. Bacteria vary in size, but a diameter of $2.0 \mu\text{m}$ is not unusual. What are the volume (in cubic centimeters) and surface area (in square millimeters) of a spherical bacterium of that size? (Consult Appendix D for relevant formulas.)

Section 1.5 Uncertainty and Significant Figures

1.11 • With a wooden ruler, you measure the length of a rectangular piece of sheet metal to be 12 mm. With micrometer calipers, you measure the width of the rectangle to be 5.98 mm. Use the correct number of significant figures: What are (a) the area of the rectangle; (b) the ratio of the rectangle's width to its length; (c) the perimeter of the rectangle; (d) the difference between the length and the width; and (e) the ratio of the length to the width?

1.12 • The volume of a solid cylinder is given by $V = \pi r^2 h$, where r is the radius and h is the height. You measure the radius and height of a thin cylindrical wire and obtain the results $r = 0.036 \text{ cm}$ and $h = 12.1 \text{ cm}$. What do your measurements give for the volume of the wire in mm^3 ? Use the correct number of significant figures in your answer.

1.13 •• A useful and easy-to-remember approximate value for the number of seconds in a year is $\pi \times 10^7$. Determine the percent error in this approximate value. (There are 365.24 days in one year.)

1.14 • Express each approximation of π to six significant figures: (a) $22/7$ and (b) $355/113$. (c) Are these approximations accurate to that precision?

Section 1.6 Estimates and Orders of Magnitude

1.15 • BIO Which of the values given below are plausible for an average adult male giraffe: (a) height of 2 m or 5 m? (b) mass of 1500 kg or 5000 kg? (c) maximal speed of 15 m/s or 45 m/s?

1.16 • How many liters of petrol are used in Italy in one day? Assume that there are two cars for every three people, that each car is driven an average of 10,000 kilometers per year, and that the petrol consumption of an average car is 8 liters per 100 kilometers. Assume that the population of Italy is approximately 60 million.

1.17 • In Wagner's opera *Das Rheingold*, the goddess Freia is ransomed for a pile of gold just tall enough and wide enough to hide her from sight. Estimate the monetary value of this pile. The density of gold is 19.3 g/cm^3 , and take its value to be about \$40 per gram.

1.18 • BIO Four astronauts are in a spherical space station. (a) If, as is typical, each of them breathes about 500 cm^3 of air with each breath, approximately what volume of air (in cubic meters) do these astronauts breathe in a year? (b) What would the diameter (in meters) of the space station have to be to contain all this air?

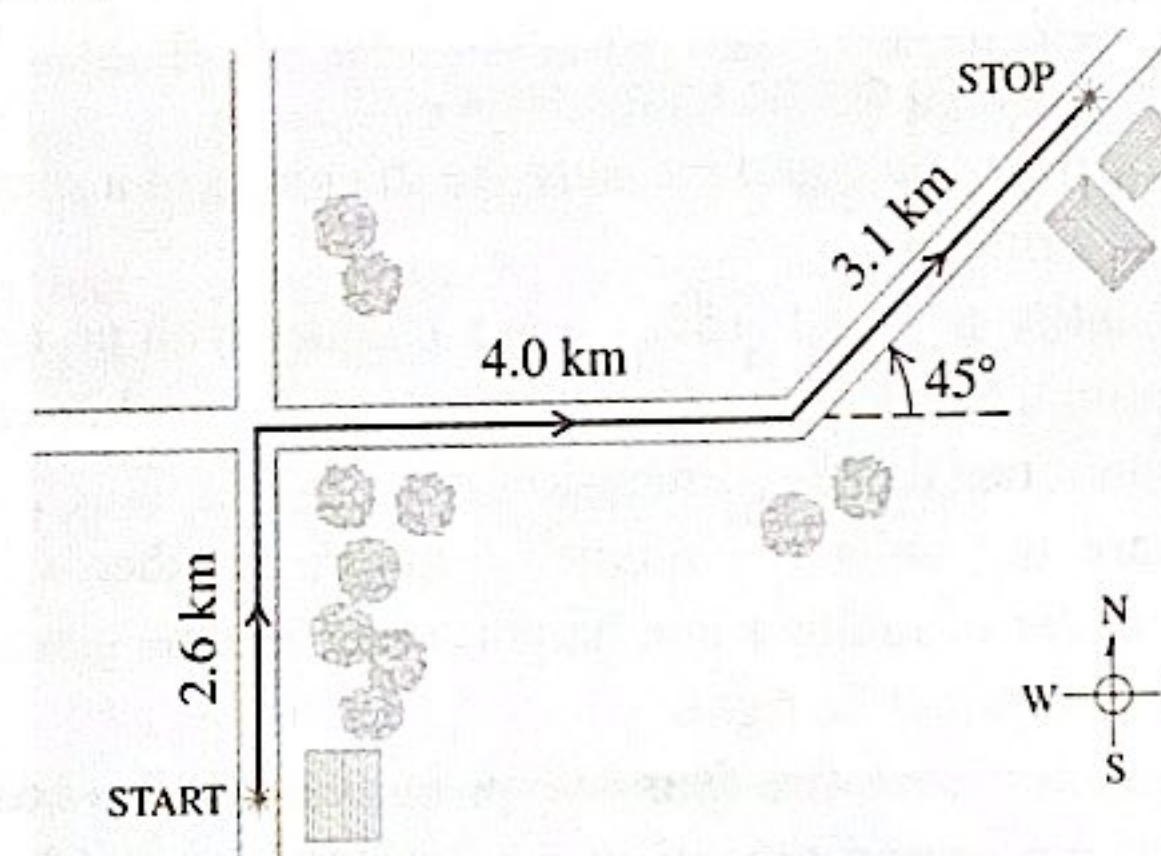
1.19 • You are using water to dilute small amounts of chemicals in the laboratory, drop by drop. How many drops of water are in a 1.0 L bottle? (Hint: Start by estimating the diameter of a drop of water.)

1.20 • BIO How many times does a human heart beat during a person's lifetime? How many liters of blood does it pump? (Estimate that the heart pumps 50 cm^3 of blood with each beat and assume a 365-day year.)

Section 1.7 Vectors and Vector Addition

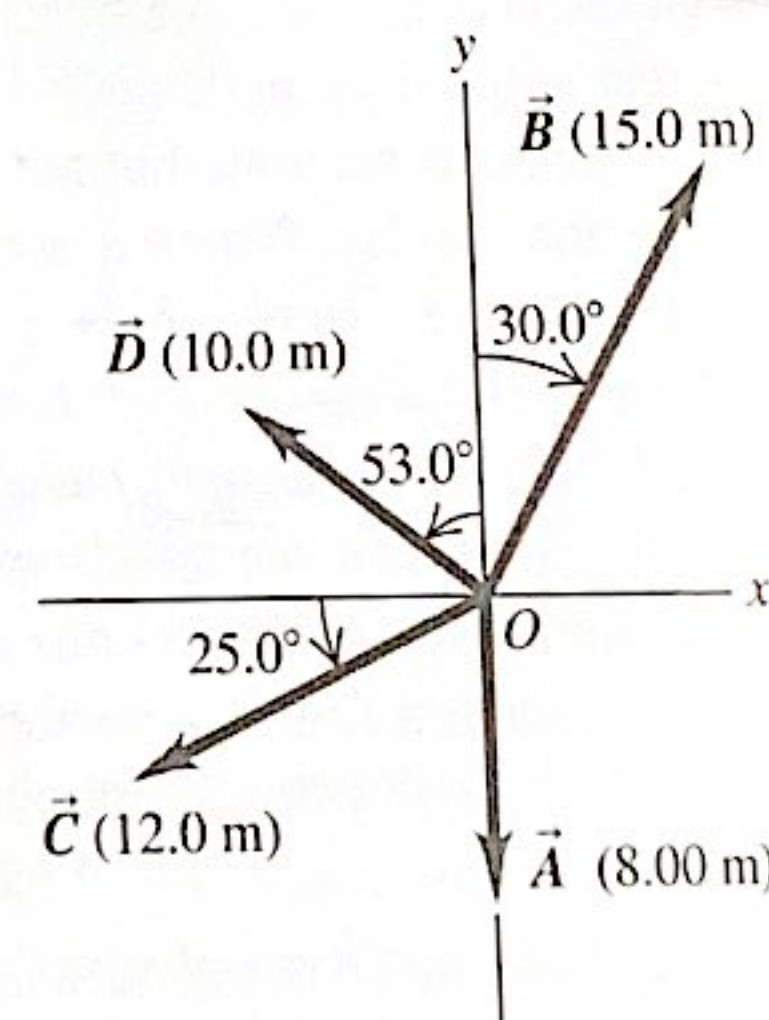
1.21 •• A postal employee drives a delivery truck along the route shown in Fig. E1.21. Determine the magnitude and direction of the resultant displacement by drawing a scale diagram. (See also Exercise 1.28 for a different approach.)

Figure E1.21



1.22 •• For the vectors $\vec{A}$ and $\vec{B}$ in Fig. E1.22, use a scale drawing to find the magnitude and direction of (a) the vector sum $\vec{A} + \vec{B}$ and (b) the vector difference $\vec{A} - \vec{B}$. Use your answers to find the magnitude and direction of (c) $-\vec{A} - \vec{B}$ and (d) $\vec{B} - \vec{A}$. (See also Exercise 1.29 for a different approach.)

Figure E1.22



1.23 •• A spelunker is surveying a cave. She follows a passage 180 m straight west, then 210 m in a direction 45° east of south, and then 280 m at 30° east of north. After a fourth displacement, she finds herself back where she started. Use a scale drawing to determine the magnitude and direction of the fourth displacement. (See also Problem 1.57 for a different approach.)

Section 1.8 Components of Vectors

1.24 •• Let θ be the angle that the vector $\vec{A}$ makes with the $+x$ -axis, measured counterclockwise from that axis. Find angle θ for a vector that has these components: (a) $A_x = 2.00 \text{ m}$, $A_y = -1.00 \text{ m}$; (b) $A_x = 2.00 \text{ m}$, $A_y = 1.00 \text{ m}$; (c) $A_x = -2.00 \text{ m}$, $A_y = 1.00 \text{ m}$; (d) $A_x = -2.00 \text{ m}$, $A_y = -1.00 \text{ m}$.

1.25 • Compute the x - and y -components of the vectors $\vec{A}$, $\vec{B}$, $\vec{C}$, and $\vec{D}$ in Fig. E1.22.